BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 14 — Hydrocarbons

ANSWER KEY (Version 1)

Total Marks: 65 | **Time Allowed: 2h 30m** | **Version: 1**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Alkanes are saturated because they contain only: √ **C — C—C single bonds** **1 mark**

Saturated hydrocarbons contain only single covalent bonds (C—C and C—H). Each carbon atom forms four bonds and each hydrogen forms only one bond. Options A and B describe unsaturated compounds (alkenes and alkynes respectively). Option D is not a hydrocarbon bond.

Q2. The general formula of alkanes is: √ **C — C_nH_{2n+2}** **1 mark**

The textbook states: "Alkanes have general formula C_nH_{2n+2}." For n=1: CH_4 (methane); n=2: C_2H_6 (ethane). Option A (C_nH_{2n}) is the formula for alkenes. Option B (C_nH_{2n-2}) is for alkynes. Option D is incorrect.

Q3. The simplest possible alkane molecule is: √ **D — Methane** **1 mark**

The textbook states: "The simplest alkane molecule that is possible is CH_4. It is called **methane**. Methane is the main component of natural gas." With only one carbon atom, it has the fewest atoms possible in an alkane. Ethane (n=2) is the next member.

Q4. Addition of hydrogen across a C=C double bond is called: √ **B — Hydrogenation** **1 mark**

The textbook (Section 14.2.1) defines: "Addition of hydrogen molecule across carbon-carbon multiple-bond is called **hydrogenation**." Halogenation involves halogens (Cl, Br). Combustion requires oxygen. Cracking breaks large molecules into smaller ones.

Q5. Catalyst used for hydrogenation at 200–300°C: √ **C — Nickel (Ni)** **1 mark**

The textbook states: "Hydrogenation takes place in presence of **finely divided nickel at 200–300°C** and high pressure." Platinum (Pt) and Palladium (Pd) can also be used but at room temperature, not 200–300°C. Iron is the catalyst for the Haber process, not hydrogenation of alkenes.

Q6. Reagent for reduction of alkyl halides producing nascent hydrogen: √ **A — Zn/HCl(aq)** **1 mark**

The textbook (Section 14.2.2) states: "Zn reacts with aqueous acid to liberate atomic hydrogen called **nascent hydrogen**." The usual reagent is Zn/HCl(aq) or Zn/H_2SO_4(aq). Sunlight is used for halogenation. Ni at high pressure is for hydrogenation. Pt/Pd is for room-temperature hydrogenation.

Q7. Breaking large hydrocarbons at 450–750°C under high pressure is called: √ **C — Cracking** **1 mark**

The textbook (Section 14.2.3): "A large hydrocarbon molecule breaks into smaller hydrocarbons when heated at temperatures such as **450–750°C and high pressure**. This process is called **thermal cracking**." Combustion requires oxygen. Hydrogenation adds H_2. Halogenation adds halogen.

Q8. Products from cracking of decane ($C_{10}H_{22}$): √ **A — Octane + Ethene** **1 mark**

The textbook gives this exact example: "$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$." Decane (10C) breaks into **octane** (8C, an alkane) and **ethene** (2C, an alkene). Note that cracking produces a mixture of alkanes and alkenes. The carbon atoms must balance: 10 = 8 + 2.

Q9. In substitution, one hydrogen atom is: √ **B — Replaced by a halogen atom** **1 mark**

The textbook (Section 14.3.1): "In a **substitution reaction** one atom or group of atoms is replaced by another atom or group of atoms." In alkane halogenation, one H on the carbon chain is replaced by one Cl (or Br). Option A describes addition, which applies to alkenes, not alkanes.

Q10. $CH_4 + Cl_2$ in diffused sunlight gives: √ **A — $CH_3Cl + HCl$** **1 mark**

The textbook gives this equation exactly: $CH_{4(g)} + Cl_{2(g)} \rightarrow \mathbf{CH_3Cl_{(g)} + HCl_{(g)}}$ in diffused sunlight. This is mono-substitution producing **chloromethane** and hydrogen chloride. Option C shows the complete substitution product CCl_4, which requires excess Cl_2 and direct sunlight.

Q11. Complete combustion of methane produces: √ **B — $CO_2 + H_2O$ + heat** **1 mark**

The textbook gives: $CH_{4(g)} + 2O_{2(g)} \rightarrow \mathbf{CO_{2(g)} + 2H_2O_{(g)}}$ **+ heat**. Complete combustion of any alkane with sufficient oxygen produces only carbon dioxide and water. Option A ($CO + H_2O$) is incomplete combustion. Option C includes carbon (soot), also incomplete combustion.

Q12. Alkanes are generally unreactive towards ionic compounds because they are: √ **B — non-polar** **1 mark**

The textbook (Section 14.3): "Alkane molecules are essentially **non-polar**. Chemically, alkanes do not react with most ionic compounds." Because alkanes have no permanent dipole, they cannot interact with ions through dipole-ion forces. This same non-polarity makes them useful as solvents for extracting non-polar substances like vegetable oils.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Definition and classification of hydrocarbons **3 marks**

Hydrocarbons: Organic compounds that contain only two elements: **carbon** and **hydrogen**. They are the simplest organic compounds. **1 mark**

Classification by bond type:

(a) **Alkanes** — single C—C bonds (saturated). Example: ethane, C_2H_6. **½**

(b) **Alkenes** — contain at least one C=C double bond (unsaturated). Example: ethene, C_2H_4. **½**

(c) **Alkynes** — contain at least one C≡C triple bond (unsaturated). Example: ethyne, C_2H_2. **1 mark**

OR

Definition of alkanes: Saturated hydrocarbons containing only single covalent bonds. Each carbon forms four bonds; each hydrogen forms one bond. **1 mark**

General formula: C_nH_{2n+2} **1 mark**

First two members:

(i) Methane: CH_4 (n = 1) — main component of natural gas.

(ii) Ethane: C_2H_6 (n = 2). **1 mark**

2(ii) — Structural formulas of methane and ethane **3 marks**

Methane (CH_4): **1 mark**

```
H | H - C - H | H
```

Ethane (C_2H_6): **1 mark**

```
H H | | H - C - C - H | | H H
```

Bond type in both: Only **single covalent bonds** (C—C and C—H). This makes them **saturated** hydrocarbons. **1 mark**

OR

Pentane (C_5H_{12}) — structural formula: **2 marks**

```
H H H H H | | | | | H - C - C - C - C - C - H | | | | | H H H H H
```

Molecular formula: C_5H_{12} (n=5: $C_5H_{2(5)+2} = C_5H_{12}$). **1 mark**

2(iii) — Hydrogenation of ethene to ethane 3 marks

Hydrogenation: The addition of hydrogen molecules (H_2) across a carbon-carbon **multiple bond** to produce a saturated compound (alkane). **1 mark**

Equation — ethene to ethane: **1 mark**

```
Ni, 200-300°C, high pressure CH2=CH2 + H2 ——————————→ CH3-CH3 Ethene Ethane
```

Catalyst: Finely divided **nickel (Ni)**. **Temperature:** 200–300°C and high pressure. (Can also use Pt or Pd at room temperature.) **1 mark**

OR

Hydrogenation of ethyne to ethene: **1.5 marks**

```
Ni, 200-300°C CH≡CH + H2 ——————————→ CH2=CH2 Ethyne Ethene
```

Why alkynes need two H_2 molecules for complete reduction to an alkane: An alkyne has a **triple bond** (C≡C), meaning it is unsaturated by two bonds. The first H_2 adds across the triple bond to give an alkene (one double bond remains). A second H_2 must then add across the double bond to give a fully saturated alkane. Each step requires one molecule of H_2. **1.5 marks**

2(iv) — Reduction of alkyl halides: chloromethane example 3 marks

Method: When an alkyl halide is treated with **Zn in the presence of an aqueous acid** (usually HCl or CH_3COOH), an alkane is produced. Zn reacts with the aqueous acid to release atomic hydrogen called **nascent hydrogen [H]**. This nascent hydrogen then reduces the alkyl halide. **1 mark**

Equation for chloromethane: **1 mark**

```
Zn/HCl(aq) CH3-Cl + 2[H] ——————————→ CH4 + HCl Chloromethane Methane
```

Nascent hydrogen: Atomic hydrogen ([H]) produced in the reaction mixture by the action of Zn on aqueous acid. It is highly reactive because it exists as single atoms, not stable H_2 molecules. **1 mark**

OR

Completion of reaction:

```
Zn/HCl(aq) CH3-CH2-Cl + 2[H] ——————————→ CH3-CH3 + HCl 1-Chloroethane Ethane
```

Product: Ethane (C_2H_6). **1.5 marks**

Reagent providing 2[H]: Zn reacts with aqueous HCl to produce nascent hydrogen: $Zn + 2HCl \rightarrow ZnCl_2 + 2[H]$. This is different from molecular H_2; nascent hydrogen is more reactive. **1 mark**

Type of reaction: Reduction — addition of nascent hydrogen to the alkyl halide is called reduction. **½**

2(v) — Cracking: definition, equation for decane, conditions 3 marks

Cracking: The process in which a large hydrocarbon (alkane) molecule is broken into smaller hydrocarbon molecules by heating at high temperatures and pressure. This is called **thermal cracking**. **1 mark**

Equation for decane: **1 mark**

```
450-750°C, high pressure C10H22 ——————————→ C8H18 + C2H4 Decane Octane Ethene
```

Products: Octane (C_8H_{18}, an alkane) and Ethene (C_2H_4, an alkene). **Conditions:** Temperature 450–750°C and high pressure. **1 mark**

OR

Industrial importance of cracking:

(i) It converts large, less useful hydrocarbons (obtained from crude oil distillation) into smaller, more useful hydrocarbons such as petrol and other fuels. **1 mark**

(ii) It produces alkenes (like ethene) which are important raw materials for making plastics, polymers, and other chemicals in the petrochemical industry. **1 mark**

Products of thermal cracking: A **mixture of alkanes and alkenes** of shorter chain length. **1 mark**

2(vi) — Physical properties of alkanes **3 marks**

(a) Polarity and solubility: Alkane molecules are essentially **non-polar**. They are **less dense than water** and do not dissolve in water. Chemically, they do not react with most ionic compounds because of their non-polar nature. **1.5 marks**

(b) States of matter based on chain length:

Alkanes containing up to **four carbon atoms** (C_1–C_4): colourless and odourless **gases**.

Alkanes containing **five to seventeen carbon atoms** (C_5–C_{17}): colourless and odourless **liquids**.

Alkanes with **more than seventeen** carbon atoms: colourless and odourless **solids**. **1.5 marks**

OR

Why hexane is a suitable solvent for vegetable oils: The textbook states: "The lack of reactivity makes them useful solvents. For example, **hexane** is used to extract vegetable oils from corn, soybeans, cotton seeds, etc." **1 mark**

Alkanes are suitable because they are: (i) **chemically unreactive** — they do not react with the oils or the plant material during extraction; (ii) **non-polar** — vegetable oils are also non-polar, so "like dissolves like" (non-polar hexane dissolves non-polar oils); (iii) easily removed by evaporation after extraction, leaving pure oil. **2 marks**

2(vii) — Substitution reaction of methane with chlorine (diffused sunlight) **3 marks**

Substitution reaction: A reaction in which one atom or group of atoms in a molecule is **replaced** by another atom or group of atoms. **1 mark**

Equation in diffused sunlight (mono-substitution): **1 mark**

```
diffused sunlight CH4(g) + Cl2(g) ————————————————→ CH3Cl(g) + HCl(g) Methane Chlorine
Chloromethane
```

Organic product: Chloromethane (CH_3Cl), also called methyl chloride. One hydrogen atom in methane is replaced by one chlorine atom. **1 mark**

OR

Reaction in direct sunlight (complete substitution): In direct (intense) sunlight, the reaction is explosive. **All four** hydrogen atoms are substituted:

```
direct sunlight CH4(g) + 2Cl2(g) ————————————→ C(s) + 4HCl(g) Methane Carbon Hydrogen chloride
```

1.5 marks

Difference: In diffused sunlight, only one H is replaced (mono-substitution), giving chloromethane. In direct sunlight, the reaction is explosive and all four H atoms are removed, producing carbon (soot) and HCl rather than an organic chloride product. The light energy is much more intense, driving the reaction to completion. **1.5 marks**

2(viii) — Photochemical reaction; why alkanes do not undergo addition **3 marks**

Photochemical reaction: A chemical reaction that is initiated or driven by the energy of **light (photons)**. The word "photo" means light. In the halogenation of alkanes, the reaction does not occur in the dark; it requires light energy to start. **1.5 marks**

The textbook states that the reaction of alkanes with halogens "takes place in presence of sunlight." Without sunlight (e.g. in the dark), CH_4 and Cl_2 do not react at room temperature. **1.5 marks**

OR

Why alkanes undergo substitution, not addition: Addition reactions require a **multiple bond** (double or triple bond) for an incoming molecule to break and attach across. Alkanes contain only **single C—C bonds** — all four valences of each carbon are already satisfied (saturated). There is no pi bond available for addition to occur. Therefore, the only way a halogen can react with an alkane is by substituting one hydrogen atom in a C—H bond for a halogen atom — a substitution reaction. **3 marks**

2(ix) — Complete combustion of methane 3 marks

Combustion: A reaction of a substance with oxygen or air that causes the **rapid release of heat** and the appearance of a flame. Complete combustion of an alkane produces carbon dioxide, water, and heat. 1 mark

Equation for complete combustion of methane: 1 mark

```
CH4(g) + 2O2(g) → CO2(g) + 2H2O(g) + heat
```

Products: Carbon dioxide (CO_2), water (H_2O), and heat. **Flame type:** Most alkanes burn with a **blue flame**. 1 mark

OR

Equation for incomplete combustion of methane: 1.5 marks

```
3CH4(g) + 4O2(g) → 2CO(g) + C(s) + 6H2O(g)
```

Three products: Carbon monoxide (CO), carbon (C, soot/black particles), and water (H_2O). 1 mark

When incomplete combustion occurs: When the supply of oxygen is **limited** (insufficient for complete combustion). This happens in poorly ventilated spaces. Carbon monoxide produced is a toxic, colourless, odourless gas. ½

2(x) — Three reasons lighter alkanes are good fuels 3 marks

The textbook (Section 14.3.2) lists three reasons why lighter alkanes are widely used as fuels:

(i) Their **combustion can be controlled** — the rate and amount of burning can be regulated, making them safe to use. 1 mark

(ii) They produce a **large amount of heat per gram** — high energy density means a small mass gives a large quantity of heat energy. 1 mark

(iii) They are **cheap and readily available** — obtained from natural gas and crude oil, which are abundant resources. 1 mark

OR

Properties of methane as a domestic fuel:

(i) **High heat output:** Complete combustion of methane ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat) releases a large amount of energy, making it efficient for cooking and heating. 1 mark

(ii) **Controllable combustion:** The gas supply can be turned on and off easily, and the rate of combustion can be adjusted using a valve. 1 mark

(iii) **Readily available and economical:** Methane is the main component of natural gas distributed through pipelines to homes, making it cheap and accessible. 1 mark

2(xi) — Preparation of propane (C_3H_8): two methods 3 marks

(a) Hydrogenation of propene: 1.5 marks

```
Ni, 200-300°C, high pressure CH3-CH=CH2 + H2 ——————→ CH3-CH2-CH3 Propene
Propane (C3H8)
```

(b) Reduction of 1-chloropropane ($CH_3CH_2CH_2Cl$): 1.5 marks

```
Zn/HCl(aq) CH3-CH2-CH2-Cl + 2[H] ——————→ CH3-CH2-CH3 + HCl 1-Chloropropane Propane (C3H8)
```

OR

(a) Hydrogenation to give ethane: 1.5 marks

```
Ni, 200-300°C CH2=CH2 + H2 ——————→ CH3-CH3 Ethene Ethane (C2H6)
```

(b) Cracking to give ethane — cracking a larger alkane such as butane: 1.5 marks

```
450-750°C, high pressure C4H10 ——————→ C2H6 + C2H4 Butane Ethane Ethene
```

(Note: Ethane appears as one of the alkane products from cracking of larger hydrocarbons.)

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Alkane structure + chemical reactivity **5 marks**

(a) Definition and structural formulas: Alkanes are saturated hydrocarbons with general formula C_nH_{2n+2}. They are called "saturated" because all carbon-carbon bonds are **single bonds** — no further atoms can be added across the bonds. Each C forms four bonds; each H forms one bond. **1 mark** **Structural formula of methane (CH_4):** **1 mark**

```
H | H - C - H | H
```

Structural formula of ethane (C_2H_6): **1 mark**

```
H H | | H - C - C - H | | H H
```

(b) Two chemical reactions of alkanes and their conditions: **2 marks**

Reaction	Condition required	Example
Halogenation (substitution)	Sunlight (photochemical)	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$
Combustion	Ignition (heat/flame) + oxygen	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat

OR

(a) Three methods of preparation of alkanes with reagents and conditions: **3 marks**

Method	Reagents	Conditions
1. Hydrogenation of alkenes/alkynes	Alkene or alkyne + H_2	Ni catalyst, 200–300°C, high pressure (or Pt/Pd at room temp)
2. Reduction of alkyl halides	Alkyl halide + Zn/HCl(aq)	Aqueous acid (HCl or CH_3COOH); produces nascent hydrogen
3. Cracking of larger hydrocarbons	Large alkane alone	450–750°C and high pressure (thermal cracking)

(b) One balanced equation per method: **2 marks**

```
Method 1 (Hydrogenation): Ni, 200-300°C CH2=CH2 + H2 ⟶ CH3-CH3 Ethene Ethane
Method 2 (Reduction of alkyl halide): Zn/HCl(aq) CH3-Cl + 2[H] ⟶ CH4 + HCl Chloromethane
Methane Method 3 (Cracking): 450-750°C, high pressure C10H22 ⟶ C8H18 +
C2H4 Decane Octane Ethene
```

3(ii) — Hydrogenation equations (ethyne to ethene to ethane) + reduction of alkyl halides **5 marks**

(a) Hydrogenation — two equations with catalyst and temperature: **3 marks** **(i) Hydrogenation of ethyne to ethene:** **1.5 marks**

```
Ni, 200-300°C CH≡CH + H2 ──────────────► CH2=CH2 Ethyne Ethene (First molecule of H2 converts the triple bond to a double bond)
```

(ii) Hydrogenation of ethene to ethane: **1.5 marks**

```
Ni, 200-300°C, high pressure CH2=CH2 + H2 ──────────────────► CH3-CH3 Ethene Ethane (Second molecule of H2 converts the double bond to a single bond) Catalyst: finely divided nickel (Ni) Temperature: 200-300°C, high pressure (Alternative: Pt or Pd catalyst at room temperature)
```

(b) Reduction of alkyl halides: **2 marks**
An alkyl halide is treated with Zn in the presence of aqueous acid. Zn reacts with HCl(aq) to liberate **nascent hydrogen [H]** (atomic hydrogen). This [H] then reduces the C—Cl bond.

```
How nascent hydrogen is produced: Zn + 2HCl(aq) ─► ZnCl2 + 2[H] (nascent hydrogen) Reduction of chloromethane: Zn/HCl(aq) CH3-Cl + 2[H] ────────► CH4 + HCl Chloromethane Methane
```

Nascent hydrogen is atomic (free) hydrogen — more reactive than molecular H_2 because it exists as single atoms. **1 mark**

OR

(a) Cracking: Thermal cracking breaks a large alkane into smaller hydrocarbons at **450–750°C and high pressure**. Large molecules from crude oil distillation are broken into more useful shorter-chain compounds. **1 mark** **Equation for cracking of decane:** **1 mark**

```
450-750°C, high pressure C10H22 ─────────────────► C8H18 + C2H4 Decane Octane Ethene
```

Temperature and pressure: 450–750°C and high pressure. Products are a **mixture of alkanes and alkenes**. **1 mark** **(b) Two industrial reasons for cracking:** **2 marks**
(i) Converts large, less volatile and less useful heavy oil fractions from crude oil into smaller, more valuable fuels like petrol (gasoline) and kerosene. **1 mark**
(ii) Produces **alkenes** (ethene, propene, etc.) which are essential raw materials for the plastics and polymer industry. Without cracking, these alkenes could not be obtained in sufficient quantities from natural sources. **1 mark**

3(iii) — Halogenation: substitution reaction + diffused vs direct sunlight **5 marks**

(a) Halogenation as photochemical substitution: **3 marks**

Substitution reaction: A reaction in which one atom or group of atoms in a molecule is **replaced by another** atom or group of atoms. In alkane halogenation, one H atom is replaced by one halogen atom (Cl or Br). **1 mark**

Photochemical reaction: The reaction is initiated by light energy (sunlight). Sunlight breaks the Cl—Cl bond, generating reactive chlorine atoms that attack the alkane. Without light, the reaction does not occur at room temperature.

Equation for mono-substitution (diffused sunlight): **1 mark**

```
diffused sunlight CH4(g) + Cl2(g) ——————————————→ CH3Cl(g) + HCl(g) Methane Chlorine
Chloromethane
```

One hydrogen atom in methane is replaced by one chlorine atom, giving **chloromethane**. **1 mark** **(b) Complete substitution in direct sunlight:** **2 marks**

In **direct sunlight**, the reaction is explosive and proceeds to completion. All four H atoms are substituted and then carbon-chlorine bonds also break:

```
direct sunlight CH4(g) + 2Cl2(g) ———————————→ C(s) + 4HCl(g) Methane Carbon Hydrogen chloride
```

Difference from mono-substitution: In diffused sunlight, only one H is replaced, yielding an organic product (chloromethane). In direct sunlight, the intense energy drives complete substitution of all H atoms and ultimately produces **carbon (soot)** and hydrogen chloride — no organic carbon compound remains. The reaction is explosive because the light energy causes a rapid chain reaction. **2 marks**

OR

(a) Physical properties of alkanes: **3 marks** **(i) Polarity:** Alkane molecules are essentially **non-polar** because carbon-hydrogen and carbon-carbon bonds have nearly equal electronegativity. There is no significant permanent dipole moment. **½** **(ii) Density compared to water:** Alkanes are **less dense than water** (density < 1.0 g/cm^3). Liquid alkanes float on water. **½** **(iii) Solubility in water:** Alkanes do **not dissolve in water**. Water is polar and "like dissolves like" — polar water cannot attract non-polar alkane molecules sufficiently to break them apart. **½** **(iv) States of matter based on chain length:** **1.5 marks**

Carbon atoms (n)	State at room temperature	Example
1 to 4	Colourless, odourless gases	Methane (CH_4), butane (C_4H_{10})
5 to 17	Colourless, odourless liquids	Hexane (C_6H_{14}), decane ($C_{10}H_{22}$)
18 and above	Colourless, odourless solids	Paraffin wax

(b) Alkanes as solvents: The lack of reactivity of alkanes with ionic compounds makes them excellent solvents for non-polar substances. Specifically, **hexane** is used to extract vegetable oils from corn, soybeans, and cotton seeds because it dissolves the non-polar oil without reacting with it and can be easily removed afterwards by evaporation. **2 marks**

3(iv) — Combustion: complete vs incomplete + alkane fuels + propane equation 5 marks

(a) Definition of combustion and comparison: Combustion is the reaction of a substance with oxygen or air that causes the rapid release of heat and appearance of a flame. ½

Feature	Complete Combustion	Incomplete Combustion
Oxygen supply	Excess (sufficient)	Limited (insufficient)
Products	CO_2, H_2O, heat	CO, C (soot), H_2O
Flame	Blue flame	Sooty / yellow flame

1.5 marks **Balanced equation — complete combustion of methane:** 1 mark

```
CH4(g) + 2O2(g) → CO2(g) + 2H2O(g) + heat
```

Balanced equation — incomplete combustion of methane: 1 mark

```
3CH4(g) + 4O2(g) → 2CO(g) + C(s) + 6H2O(g)
```

(b) Three reasons lighter alkanes make good fuels: 1.5 marks
(i) Their combustion can be **controlled**. ½
(ii) They produce a **large amount of heat per gram**. ½
(iii) They are **cheap and readily available** from natural gas and oil. ½
Combustion of propane (C_3H_8):

```
C3H8(g) + 5O2(g) → 3CO2(g) + 4H2O(g) + heat
```

OR

(a) Preparation of propane (C_3H_8) by three methods: 3 marks **(i) Hydrogenation of propene:** 1 mark

```
Ni, 200-300°C, high pressure CH3-CH=CH2 + H2 ————————————→ CH3-CH2-CH3 Propene Propane (C3H8)
```

(ii) Reduction of 1-chloropropane: 1 mark

```
Zn/HCl(aq) CH3-CH2-CH2-Cl + 2[H] ——————————→ CH3-CH2-CH3 + HCl 1-Chloropropane Propane (C3H8)
```

(iii) Cracking a larger alkane: For example, cracking pentane (C_5H_{12}): 1 mark

```
450-750°C, high pressure C5H12 ————————————→ C3H8 + C2H4 Pentane Propane Ethene
```

(b) Photochemical halogenation of ethane: 2 marks
Ethane reacts with chlorine in **diffused sunlight** via a substitution reaction. One hydrogen atom on an ethane molecule is replaced by one chlorine atom:

```
diffused sunlight CH3-CH3 + Cl2 ————————————→ CH3-CH2-Cl + HCl Ethane Chlorine Chloroethane
```

Organic product: Chloroethane (CH_3CH_2Cl), also called ethyl chloride. The reaction is photochemical because it requires light energy to proceed. 2 marks